Unit Test Report

Field	Detail
Project	Vehicle Parking System (VPS) v1.1 — Group 6
Generated	11 Nov 2026, 08:38 IST
Environment	Python 3.13.18 · Linux x86_64
Source revision	84bf5b3
Prepared by	Himansh Ahuja (QA Lead) · generated by scripts/generate_reports.py

1. Summary

Metric	Value
Test files	18
Test cases executed	**221**
Passed	**221**
Failed / errors	0
Skipped	0
Pass rate	**100.0 %**
Execution time	17.69 s
Statement coverage (unit tests only)	99.8 %
Branch coverage (unit tests only)	98.7 %

Verdict: all unit tests pass.

Scope: every module in `vps/` is tested in isolation against an in-memory database and a controllable clock (`tests/conftest.py`). Hardware is replaced by the simulators in `vps/hardware.py`. Each test is tagged with the Test Case ID(s) it implements, so results trace back to the Test Case Specification.

2. Results by test file

Test file	Owner	Tests	Passed	Failed	Skipped	Time (s)
`tests/unit/test_admin_api.py`	Himansh	8	8	0	0	4.43
`tests/unit/test_allocation.py`	Shweta	14	14	0	0	0.13
`tests/unit/test_app.py`	Angelo	9	9	0	0	1.53
`tests/unit/test_audit.py`	Angelo	5	5	0	0	0.01
`tests/unit/test_auth.py`	Angelo	15	15	0	0	6.17
`tests/unit/test_edge.py`	Shweta	4	4	0	0	0.01
`tests/unit/test_edge_api.py`	Shweta	1	1	0	0	0.48
`tests/unit/test_egress.py`	Akarsh	15	15	0	0	0.76
`tests/unit/test_hardware.py`	Shweta	20	20	0	0	0.01

Test file	Owner	Tests	Passed	Failed	Skipped	Time (s)
`tests/unit/test_ingress.py`	Angelo	22	22	0	0	1.42
`tests/unit/test_payment.py`	Akarsh	10	10	0	0	0.03
`tests/unit/test_receipt.py`	Akarsh	4	4	0	0	0.09
`tests/unit/test_reports.py`	Himansh	20	20	0	0	0.80
`tests/unit/test_reservation.py`	Himansh	14	14	0	0	0.78
`tests/unit/test_security.py`	Angelo	24	24	0	0	0.33
`tests/unit/test_sensors.py`	Shweta	11	11	0	0	0.59
`tests/unit/test_tariff.py`	Akarsh	23	23	0	0	0.07
`tests/unit/test_tls.py`	-	2	2	0	0	0.04
Total		**221**	**221**	**0**	**0**	**17.69**

3. Test cases exercised by unit tests

Test case	Unit tests
TC-ADM-01	6
TC-ADM-02	19
TC-BILL-01	10
TC-BILL-02	5
TC-BILL-03	7
TC-ENT-01	9
TC-ENT-02	3
TC-ENT-03	5
TC-ENT-04	5
TC-ENT-05	1
TC-ENT-06	5
TC-Fault-01	5
TC-PAY-01	1
TC-PAY-02	2
TC-PAY-03	2
TC-PAY-04	4
TC-PAY-05	3
TC-RES-01	5
TC-RES-02	2
TC-RES-03	4
TC-SEC-01	4
TC-SEC-02	10
TC-SEC-03	22

Test case	Unit tests
TC-SEC-04	9
TC-SEC-05	8
TC-SEC-06	1
TC-SLOT-01	6
TC-SLOT-02	2
TC-SLOT-03	2
TC-SLOT-04	3

4. Failures

None.

5. Slowest tests

Test	Time (s)
`test_auth::test_wrong_password_and_lockout`	0.880
`test_admin_api::TestOverrideAndAudit::test_audit_verify_endpoint`	0.676
`test_admin_api::TestTariffAdmin::test_update_tariff_live`	0.655
`test_reports::test_reports_api`	0.646
`test_app::test_rbac_guards`	0.585
`test_app::test_auth_api`	0.585
`test_admin_api::TestTariffAdmin::test_surge_and_coupons`	0.581
`test_admin_api::TestOverrideAndAudit::test_override_is_audited`	0.569
`test_egress::TestEgressApi::test_cash_endpoint_and_sandbox_disabled_in_prod`	0.567
`test_ingress::test_api_manual_entry_requires_staff`	0.536

6. Artefacts

- Interactive HTML report: `docs/reports/html/unit-tests.html`
- Raw JUnit XML is produced by CI (`junit.xml` build artefact)
- Re-run: `pytest tests/unit -v` or `python scripts/generate_reports.py`